

3.4 Consecutive Angles

Lesson 4 Introduction

 Benchmark	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.		 Learning Targets	- I can determine the relationships of consecutive exterior and interior angles. - I can solve problems that involve angles formed by parallel lines intersected by a transversal.
 Benchmark Coherence	Building On MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.		Working Toward N/A	
 Essential Questions	- What relationship do consecutive interior angles have? - What relationship do consecutive exterior angles have? - How can you use angle relationships to find unknown values and unknown angle measures?			
 Lesson Narrative	In this lesson, the angle relationships of consecutive interior angles and consecutive exterior angles created by two parallel lines intersected by a transversal are established. Practice includes using these relationships along with the angle relationships introduced in the prior lessons to solve for unknown values and unknown angle measures.			
 Component Summary	3.4.1 Warm-Up (~5 min.)	Connect numbers with their factors through area (<i>SE p. 144</i>)	3.4.3 Guided Practice (10 min.)	Determine values of unknowns and angle measures using angle relationships (<i>SE p. 148</i>)
	3.4.2 Collaboration (25-30 min.)	Prove the relationships of Consecutive Exterior Angles and Consecutive Interior Angles Theorems (<i>SE p. 145</i>)	3.4.4 Wrap-Up (~5 min.)	Reflect on confidence with angle relationships explored in the unit (<i>SE p. 149</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Consecutive Interior Angles Theorem - Consecutive Exterior Angles Theorem <u>Additional Terms:</u> N/A		(Optional) Patty Paper	Consider providing a word bank for students to choose from when completing the proofs in 3.4.2 Collaboration.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse consecutive interior and exterior angles with corresponding angles or alternate interior or exterior angles. - Students may relate an angle measure to angles created with the wrong transversal.	Students may get the various angle definitions mixed up, either because they sound similar, or because they have similar locations. Consider providing a Word Wall with the definitions and pictures to support students as they learn this academic vocabulary.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time In this lesson, students are provided with an opportunity to refine their verbal and written responses with questions found in the 3.4.2 Collaboration. Students will discuss with partners and others in the class to compare and revise their conjectures.	MA.K12.MTR.1.1: Participate in Effortful Learning 3.4.2 Collaboration provides an opportunity for students to analyze an image to discover the relationships of consecutive interior and exterior angles. Encourage students to help and support each other when attempting a new method or approach. Students also build their perseverance in 3.4.3 Guided Practice by completing questions involving multiple transversals and angle measures represented by algebraic expressions.
 Differentiate Instruction	Consider providing patty paper in this lesson so that students can more visually see the congruence of angles, and that consecutive interior and exterior angles sum to 180° . Word Walls or word banks can also be used for scaffolded support of students when completing proofs and describing angle relationships in 3.4.2 Collaboration and 3.4.3 Guided Practice.	
3.4.2 Collaboration allows students to discuss and correct their answers through auditory and visual entry points when discovering the relationships of consecutive interior and consecutive exterior angles.		
Lesson Notes:		

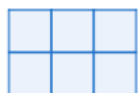
3.4.1 Warm-Up: Squares in Rectangles

Component Overview: Students make the connection between the dimensions of a rectangle and its area and how the possible dimensions are the factors of the area of the rectangles.

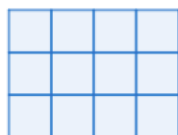


3.4.1 Warm-Up: Squares in Rectangles

Three rectangles and their descriptions are shown.



A 2×3 rectangle contains 6 unit squares.



A 3×4 rectangle contains 12 unit squares.



A 4×6 rectangle contains 24 unit squares.

- What are the dimensions of a rectangle that contains exactly 100 unit squares?
A 10×10 square.
- List any other size rectangles that contain exactly 100 unit squares.
Answers will vary.
 1×100 ; 2×50 ; 4×25 ; 5×20



For students who finish early, consider extending these questions by having students consider the following statement and defend it as always, sometimes, or never true:

“The larger the number of squares a rectangle has, the more options there are for the dimensions of the rectangle.”

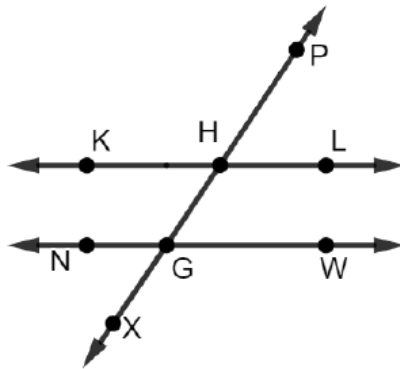
3.4.2 Collaboration: Consecutive Interior and Exterior Angles

Component Overview: Students use an image of parallel lines intersected by a transversal to determine the congruence of pairs of angles and their angle pair name based on their location in the image. Students then write a conjecture about the relationships of consecutive interior angles and consecutive exterior angles, verifying their conjectures by completing a two-column proof before applying the relationships to solve questions.



3.4.2 Collaboration: Consecutive Interior and Exterior Angles

1. $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$ where $\overleftrightarrow{XP}$ is a transversal.



- a. Work with your partner to determine if each pair of angles could be congruent by using corresponding, alternate exterior, alternate interior, or vertical angles.

Angles	Congruent?
$\angle NGH$ and $\angle KHG$	<i>NOT congruent</i>
$\angle PHL$ and $\angle WGX$	<i>NOT congruent</i>

- b. Work with your partner to determine which pair of angles in the table could be described as consecutive interior angles.
 $\angle NGH$ and $\angle KHG$ are consecutive interior angles.
- c. Name another pair of consecutive interior angles.
 $\angle WGH$ and $\angle LHG$ are also consecutive interior angles.



3.4.2 Collaboration is scaffolded to allow students to engage with it with their partner or small group, rather than being guided through by the teacher. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overhead while they collaborated for the whole class to hear, or to address common misconceptions that were evident among the majority of the class.



Consider having a Word Wall or poster with the various angle relationships that have been taught in the last few lessons, including a visual of each. This will allow students to have a reference point when working through this component with their partner.



As students start questions 1b and 1c, suggest having students breakdown the term “consecutive interior angles” to determine a description of the angle pairs prior to naming the angles.

3.4.2 Collaboration: Consecutive Interior and Exterior Angles (continued)

- d. Work with your partner to write a conjecture about the relationship that is true for consecutive interior angles.

Conjecture: Consecutive interior angles are supplementary.

- e. Ask another classmate for their conjecture about consecutive interior angles. Record their conjecture and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials
<i>Answers will vary.</i>		

- f. Compare the conjecture from your classmate with the conjecture you and your partner wrote. If necessary, make any adjustments to your conjecture.

Answers will vary.

- g. $\angle PHL$ and $\angle WGX$ are consecutive exterior angles. Name another pair of consecutive exterior angles.

$\angle XGN$ and $\angle PHK$ are consecutive exterior angles.

- h. Work with your partner to write a true conjecture about the relationship between consecutive exterior angles.

Conjecture: Consecutive exterior angles are supplementary.



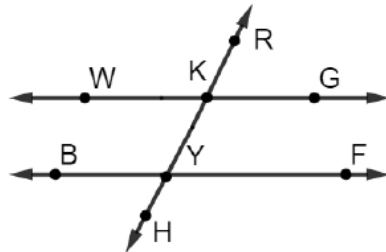
What are the differences and similarities between consecutive interior and consecutive exterior angles?

3.4.2 Collaboration: Consecutive Interior and Exterior Angles (continued)

2. A partially completed proof is shown.

a. Work with your partner to complete the proof.

Given: $\overline{WG} \parallel \overline{BF}$
 Prove: $\angle WKY$ and $\angle BYK$ are supplementary.



Statement	Reason
1. $\overline{WG} \parallel \overline{BF}$	1. Given
2. $\angle WKY \cong \angle BYH$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle BYH$ and $\angle BYK$ are a linear pair	3. Definition of a linear pair
4. $\angle BYH$ and $\angle BYK$ are supplementary	4. If two angles form a linear pair, then they are supplementary.
5. $m\angle BYH + m\angle BYK = 180^\circ$	5. Definition of supplementary angles
6. $m\angle WKY = m\angle BYH$	6. Definition of congruence
7. $m\angle WKY + m\angle BYK = 180^\circ$	7. Substitution
8. $\angle WKY$ and $\angle BYK$ are supplementary.	8. Definition of supplementary angles

b. Check your proof with another classmate. If necessary, make any adjustments.



Consider providing a word bank for students who are struggling to fill out the proof.

Below is a possible word bank to use. Angles can be used more than once.

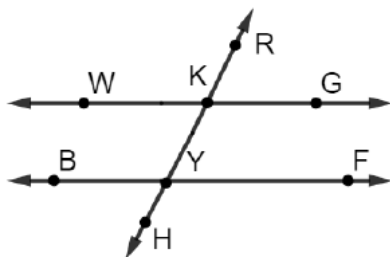
- BYH
- WKY
- $\angle BYH$
- $\angle WKY$
- $m\angle BYH$
- $m\angle WKY$
- Definition of congruence
- Definition of supplementary angles
- Definition of complementary angles
- $\angle BYH$ and $\angle BYK$ are supplementary
- $\angle BYH$ and $\angle FYH$ are supplementary

3.4.2 Collaboration: Consecutive Interior and Exterior Angles (continued)

3. Work with your partner to complete the proof.

Given: $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$

Prove: $\angle RKG$ and $\angle FYH$ are supplementary.



Statement	Reason
1. $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$	1. Given
2. $\angle RKG \cong \angle KYF$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle KYF$ and $\angle FYH$ are a linear pair	3. Definition of a linear pair
4. $\angle KYF$ and $\angle FYH$ are supplementary	4. If two angles form a linear pair, then they are supplementary.
5. $m\angle KYF + m\angle FYH = 180^\circ$	5. Definition of supplementary angles
6. $m\angle RKG = m\angle KYF$	6. Definition of congruence
7. $m\angle RKG + m\angle FYH = 180^\circ$	7. Substitution
8. $\angle RKG$ and $\angle FYH$ are supplementary.	8. Definition of supplementary angles



Consider providing a word bank for students who are struggling to fill out the proof.

Below is a possible word bank to use. Angles can be used more than once.

- KYF
- RKG
- $\angle KYF$
- $\angle RKG$
- $m\angle KYF$
- $m\angle RKG$
- $m\angle KYF + m\angle KYB = 180^\circ$
- $m\angle KYF + m\angle FYH = 180^\circ$
- $m\angle RKG \cong m\angle BYH$
- $m\angle RKG \cong m\angle KYF$
- Definition of congruence
- Definition of supplementary angles
- Definition of complementary angles
- $\angle BYH$ and $\angle KYB$ are supplementary
- $\angle KYF$ and $\angle FYH$ are supplementary



If students are creating a proof resource booklet or guide, consider having them add common statements that “typically” follow each other in a proof. For example, the following statements typically, but not always, follow each other in a proof:

$\underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}}$ are linear pairs definition of linear pair
 $\underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}}$ are supplementary linear pair postulate
 $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 180^\circ$ definition of supplementary angles

3.4.2 Collaboration: Consecutive Interior and Exterior Angles (continued)

4. Mistakes can be an opportunity to learn.

- a. What mistakes did you make in the first proof that helped you in the second proof?

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

- b. If you did not make a mistake in the first proof, describe how the proofs were similar.

Answers will vary.



Consecutive Interior Angles Theorem

If two parallel lines are intersected by a transversal, then interior angles on the same side of the transversal are supplementary.



Consecutive Exterior Angles Theorem

If two parallel lines are intersected by a transversal, then exterior angles on the same side of the transversal are supplementary.



Consider having students compare their mistakes with their partner and discuss with each other how to learn from them to prevent making them in the future.

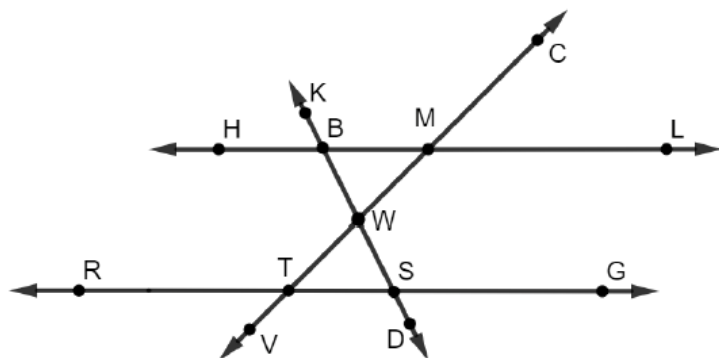
3.4.3 Guided Practice: Angles in Parallel Lines

Component Overview: Students apply relationships of angles formed by two parallel lines intersected by a transversal to solve for unknown values and angle measures.



3.4.3 Guided Practice: Angles in Parallel Lines

1. $\overline{HL} \parallel \overline{RG}$ where $\overline{CV}$ and $\overline{KD}$ are transversals with $m\angle KBM = 117^\circ$ and $m\angle VTS = 134^\circ$.



Determine the measurement of each angle.

$$m\angle DSG = 63^\circ$$

$$m\angle LMC = 46^\circ$$

$$m\angle RTV = 46^\circ$$

$$m\angle BWM = 71^\circ$$



Allow informal observation throughout the first two components to dictate how “guided” 3.4.3 Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



Consider having students either write or verbalize a justification for how they found each angle measurement. This justification would prevent random guessing on the answer.

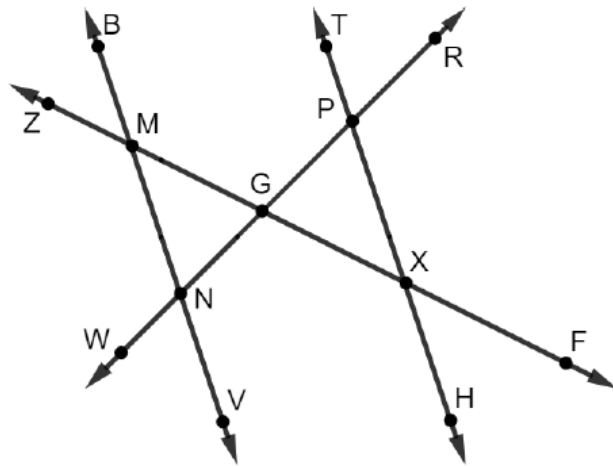
An example of a justification could be:

$m\angle DSG = 63^\circ$ because it is a consecutive exterior angle, and supplementary to $\angle KBM$ formed by parallel lines $\overline{HL}$ and $\overline{RG}$, intersected by transversal $\overline{KD}$.

Having the students name the transversal when two transversals are shown will help to clear misconceptions students have about angle relationships.

3.4.3 Guided Practice: Angles in Parallel Lines (continued)

2. In the figure shown, $\overleftrightarrow{BV} \parallel \overleftrightarrow{TH}$, $m\angle GXH = (5(2x - 1))^\circ$, $m\angle GMN = (4x - 11)^\circ$, $m\angle XPG = (3y - 8)^\circ$, and $m\angle VNG = (7y - 52)^\circ$.



- a. Find the value of x .
 $(4x - 11) + (5(2x - 1)) = 180$
 $x = 14$
- b. Find the value of y .
 $(7y - 52) + (3y - 8) = 180$
 $y = 24$



Before having students start on question 2, consider having them name the angle relationships of the given angles along with the transversal.

This support will allow students who are still confused on how to apply the angle relationships, and provides a way for them to map out their thinking.

3.4.4 Wrap-Up: Reflection

Component Overview: Students reflect on their confidence and understanding of the angle relationships explored in Lessons 1 – 4.



3.4.4 Wrap-Up: Reflection

1. Rate your confidence in solving problems using these angle pairs.

Answers will vary.

I can solve number problems involving....	I am an expert and could help others.	I can hold my own.	I am getting there.	I would like more help or practice.
Congruent Supplements				
Congruent Complements				
Vertical Angles				
Consecutive Interior Angles				
Consecutive Exterior Angles				
Alternate Interior Angles				
Alternate Exterior Angles				
Corresponding Angles				



What would reflection look like for you as a teacher? Consider your own informal data regarding students' understanding of these angle relationships thus far in the unit and how best to address potential gaps in understanding in upcoming lessons.